4. Operating Modes

4.1 Overview

Table 4.1 shows the selection of operating modes by the mode-setting pin(MD pin) and JTAG accessible command. For details, see section 4.3. Details of Operating Modes.

Table 4.1 Selection of operating modes by the mode-setting pin

Mode-setting pin (MD)	Operating mode	On-chip MRAM	External bus
1	Single chip mode/ JTAG Boot Mode	Enable	Disable
0	SCI / USB boot mode	Enable	Disable

4.2 Operating Mode Types and Selection

Table 4.1 show the relationship between levels on the mode-setting pins (MD) on release from the reset state and the operating mode selected at that time. For details on each of the operating modes, see section 4.3. Details of Operating Modes. Operation starts with the on-chip MRAM enabled and the external bus disabled, regardless of the mode in which operation started.

4.3 Details of Operating Modes

4.3.1 Single-Chip Mode

In single-chip mode, all I/O pins are available for use as input or output port, inputs or outputs for peripheral functions, or as interrupt inputs. When a reset is released while the MD pin is high, the MCU starts in single-chip mode and the on-chip MRAM is enabled.

4.3.2 JTAG Boot Mode

In this mode, the on-chip MRAM programming routine (JTAG boot program), stored in a dedicated area within the MCU, is used. The on-chip MRAM, including code MRAM and extra MRAM, can be modified from outside the MCU by using the JTAG and SWD interface. CSW.DbgSwEnable bit in APB-AP Control/Status Word register must be 1 in this mode.

To enter this mode, it is necessary to input the request from the JTAG and SWD-I/F during RES-pin reset.

4.3.3 SCI Boot Mode

In this mode, the on-chip MRAM programming routine (SCI boot program), stored in a dedicated area within the MCU, is used. The on-chip MRAM, including the code MRAM and extra MRAM, can be modified from outside the MCU by using a universal asynchronous receiver/transmitter (UART) SCI. For details, see section 58, MRAM. The MCU starts in SCI boot mode if the MD pin is held low on release from the reset state.

4.3.4 USB Boot Mode

In this mode, the on-chip MRAM programming routine (USB boot program), stored in a dedicated area within the MCU, is used. The on-chip MRAM, including the code MRAM and extra MRAM, can be modified from outside the MCU by using the USB. For details, see section 58, MRAM. The MCU starts in USB boot mode if the MD pin is held low on release from the reset state.

4.4 Operating Modes Transitions

4.4.1 Operation Modes and the Relationship of Mode Transition

Figure 4.1 shows that the operation modes and the relationship of mode transition. Operating mode can be transitioned in the direction of the arrow at the figure.

JTAG Boot mode and SCI/USB Boot mode is not possible using POR.

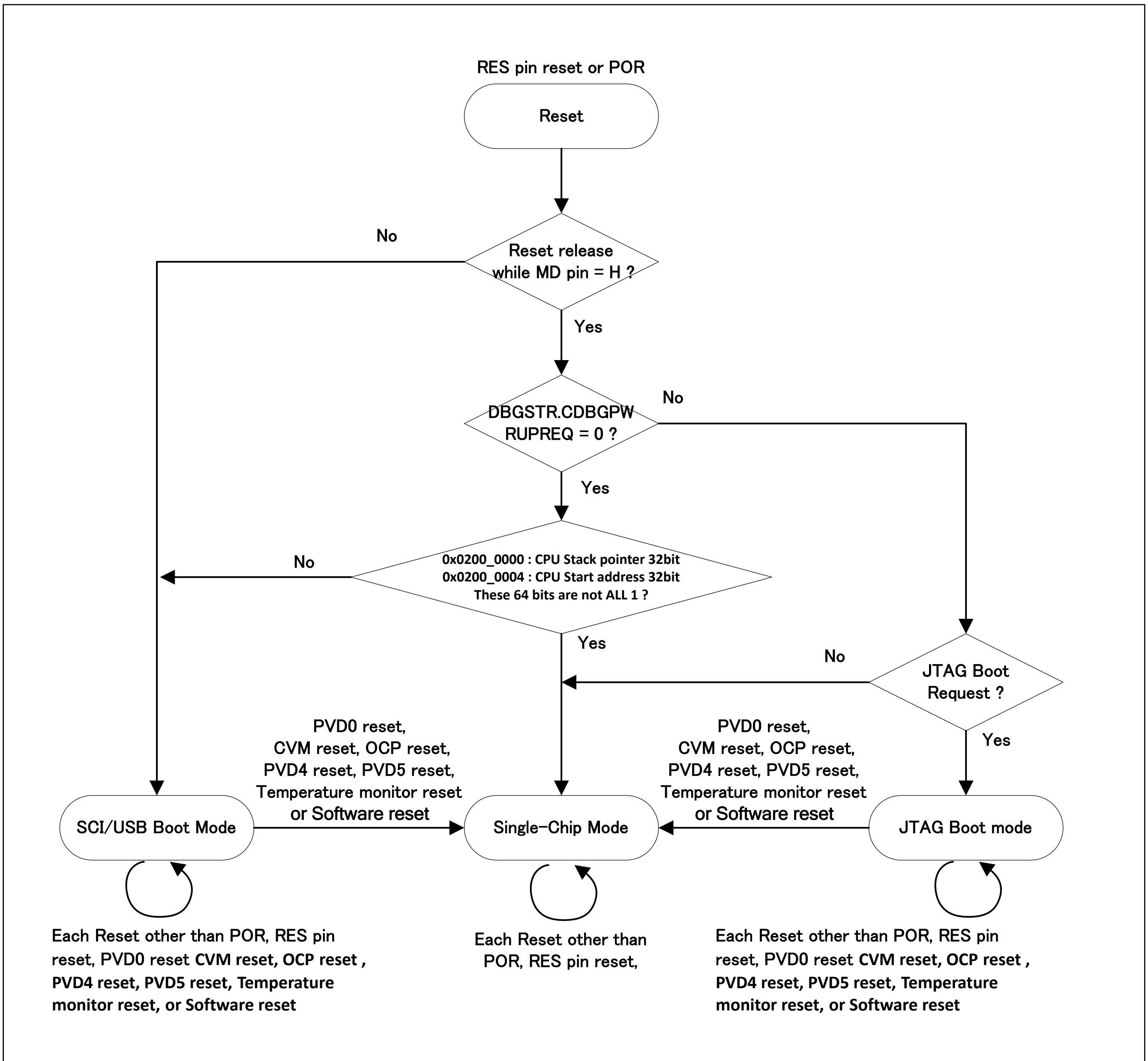

Figure 4.1 Operation modes and the relationship of mode transition